

Pre-Primary Tutor Using Arduino



K. SITARAMA RAO

A simple tutor for kids is presented here. An Arduino board (or its equivalent Free-duino), three tactile switches and a 16x2 LCD module are the main components used here.

Arduino is selected because it is an open source prototyping electronics platform based on flexible, easy-to-use hardware and software. It is intended for designing creative applications by hobbyists, artists and engineers.

Circuit and working

The circuit diagram of the pre-primary tutor for kids is shown in Fig. 1. ATmega8 based board is used in this design.

Software is written in Arduino programming language, which is similar to C language. Software is written and compiled using open source Arduino integrated development environment (IDE). It can be downloaded from arduino.cc. Language structure, description and library functions are also given at arduino.cc. The compiled code is also uploaded on Arduino through Arduino IDE.

Switches S1, S2 and S3 are connected to digital pins 8, 9 and 10 of Arduino, respectively. These digital pins are programmed as input pins in set-up function of sketch. Programs written for Arduino are called sketches. The set-up function in sketch is used to set up pin modes, serial ports, etc. The actual functional code will be in the loop function of sketch.

When a switch is in open condition, the corresponding digital pin is connected to ground through pull-down resistors of value 10-kilo-ohm each. When a switch is pressed, the corresponding digital pin is con-

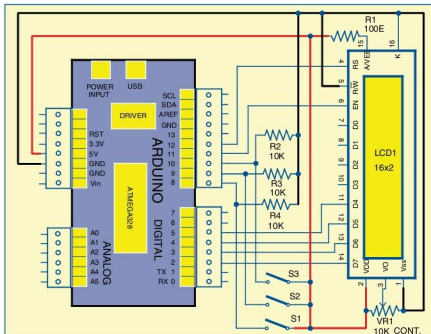


Fig. 1: Circuit diagram of the pre-primary tutor for kids

TABLE I
LCD Display in Mode 0

	Row 1	Row 2
Pressing S1 repeatedly	WELCOME	A a, B b, ..., Z z in a sequence
Pressing S2 repeatedly	WELCOME	A Apple, B Boy, ..., Z Zebra in a sequence

TABLE II
LCD Display in Mode 1

	Row 1	Row 2
Pressing S1 repeatedly	WELCOME	Increments multiplier value (a) from 1 to 20 in a sequence
Pressing S2 repeatedly	WELCOME	Increments multiplicand value (b) from 1 to 20 in a sequence Displays multiplication tables as $a \times b = c$

TABLE III
LCD Display in Mode 2

	Row 1	Row 2
Pressing S1 repeatedly	WELCOME	Sunday, Monday, ..., Saturday in a sequence
Pressing S2 repeatedly	January, February, ..., December in a sequence	31 days, 28/29 days, ..., 31 days; corresponding number of days for every month in a sequence